Python Project Topic

Project Title: **Library Management System**

I'll create a library management system where I'll utilize a three-level hierarchy of classes for different types of library items (e.g., books, magazines, DVDs). Each item will inherit from a base class representing a library Item. The program will have functions that are polymorphic, such as a function to calculate late fees, which might differ for different types of items.

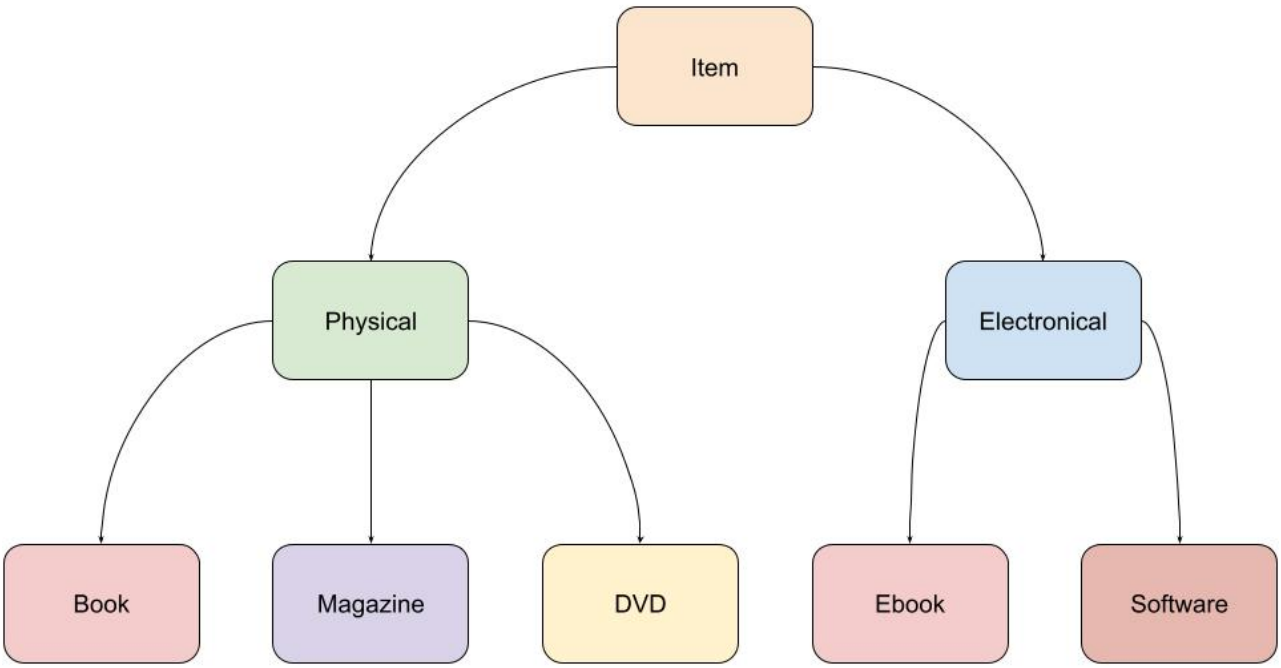
Features:

- **Input files:** Load initial data such as book titles, authors, publication dates, etc., from input file(s). It will be either one of the two options:
    - One file with all the necessary information.
    - Several files that represent a different type of library item.
- **Data Structures:** I will use data structures like arrays, linked lists, or binary trees to organize collections of library items. It might have a central library class that contains these collections, each storing objects of different types (e.g., books, magazines).
- **Algorithms:** Implementing algorithms to manipulate collections of library items. For example, I might implement an algorithm to search for a specific item by title or author, or to sort items based on different criteria such as publication date or popularity.
- **Effective algorithms:** I will avoid using $O(N^2)$ algorithms like bubble sort. Instead, I will opt for algorithms with better time complexity such as quicksort, merges ort, or binary search – I haven't decided yet.
- **Filtering:** A function to list all products within a specific price range.
- **Error handling**: Handle various errors gracefully, such as attempting to check out an item that is already checked out, or searching for a non-existent item.
- **Unit Testing:** Test cases, to check that error handling is working, and everything shows no problems.
    - Examples: testing that *calculate_late_fee()* returns the correct amount
      physical_condition() is valid
      code_expiration() is a correct date
      is not expired

System components:

- *Item* (Abstract/Base class)
    - Functions: c*alculate_late_fee()* [polymorphic]
- *Physical (Item)*
    - Functions: p*hysical_condition()*

- *Electronical (Item)*
    - Functions: *code_expiration()*
- *Book* (*Physical)*
    - Attributes (non-public members): *author, genre, year*
- *E-book (Electronical)*
    - Attributes (non-public members): a*uthor, genre, year*
- *Magazine* (*Physical)*
    - Attributes (non-public members): *issue_number, publisher*
- *DVD* (*Physical)*
    - Attributes: (non-public members) *director, genre, year, duration*
- *Software* (*Electronical)*
    - Attributes (non-public members): *username, subscription_plan, usage_time*

Take in mind that more features (e.g. classes, functions) might be added. Therefore, this is not the final take on the components.

This is the hierarchy of my project: